

2019 年数学(三)真题解析

一、选择题

(1) 【答案】 (C).

【解】 方法一 由 $\lim_{x \rightarrow 0} \frac{x - \tan x}{x^3} = \lim_{x \rightarrow 0} \frac{1 - \sec^2 x}{3x^2} = -\frac{1}{3}$ 得 $x - \tan x \sim -\frac{1}{3}x^3 (x \rightarrow 0)$,

故 $x - \tan x$ 为 3 阶无穷小, 即 $k=3$, 应选(C).

方法二 由 $\tan x = x + \frac{1}{3}x^3 + o(x^3)$ 得 $x - \tan x \sim -\frac{1}{3}x^3 (x \rightarrow 0)$,

故 $k=3$, 应选(C).

(2) 【答案】 (D).

【解】 令 $f(x) = x^5 - 5x + k$,

则 $f'(x) = 5x^4 - 5 = 5(x^2 + 1)(x^2 - 1)$,

当 $x < -1$ 时, $f'(x) > 0$; 当 $-1 < x < 1$ 时, $f'(x) < 0$; 当 $x > 1$ 时, $f'(x) > 0$,

极大值 $f(-1) = 4 + k$, 极小值 $f(1) = -4 + k$,

又 $\lim_{x \rightarrow -\infty} f(x) = -\infty$, $\lim_{x \rightarrow +\infty} f(x) = +\infty$,

故方程 $x^5 - 5x + k = 0$ 有 3 个实根的充分必要条件是 $f(-1) > 0, f(1) < 0$, 即 $-4 < k < 4$, 应选(D).

(3) 【答案】 (D).

【解】 微分方程 $y'' + ay' + by = ce^x$ 的特征方程为 $\lambda^2 + a\lambda + b = 0$,

由 $y = (C_1 + C_2x)e^{-x} + e^x$ 为方程的通解可知, 其特征根 $\lambda_1 = \lambda_2 = -1$,

则 $a = 2, b = 1$;

再由 $y^* = e^x$ 为微分方程 $y'' + ay' + by = ce^x$ 的特解得 $c = 4$, 应选(D).

(4) 【答案】 (B).

【解】 因为级数 $\sum_{n=1}^{\infty} \frac{v_n}{n}$ 条件收敛, 所以 $\lim_{n \rightarrow \infty} \frac{v_n}{n} = 0$,

则存在 $M > 0$, 对一切的 n , 有 $\left| \frac{v_n}{n} \right| \leq M$;

而 $0 \leq |u_n v_n| = \left| \frac{v_n}{n} \right| \cdot |nu_n| \leq M |nu_n|$, 且 $\sum_{n=1}^{\infty} M |nu_n|$ 收敛,

由正项级数的比较审敛法得 $\sum_{n=1}^{\infty} |u_n v_n|$ 收敛, 即 $\sum_{n=1}^{\infty} u_n v_n$ 绝对收敛, 应选(B).

(5) 【答案】 (A).

【解】 因为 $AX = 0$ 的基础解系中含 2 个解向量, 所以 $r(A) = 2 < 4$,

故 $r(A^*) = 0$, 应选(A).

(6) 【答案】 (C).

【解】 令 $AX = \lambda X (X \neq 0)$,

由 $A^2 + A = 2E$ 得 $(A^2 + A - 2E)X = (\lambda^2 + \lambda - 2)X = 0$,

从而有 $\lambda^2 + \lambda - 2 = 0$, 即 $\lambda = -2$ 或 $\lambda = 1$,

因为 $|\mathbf{A}|=4$, 所以 $\lambda_1=1, \lambda_2=\lambda_3=-2$,

故二次型 $\mathbf{X}^T \mathbf{A} \mathbf{X}$ 的规范形为 $y_1^2 - y_2^2 - y_3^2$, 应选(C).

(7) 【答案】 (C).

【解】 由减法公式得 $P(\overline{AB}) = P(A) - P(AB)$, $P(\overline{BA}) = P(B) - P(AB)$,

则 $P(A) = P(B)$ 的充分必要条件是 $P(A) - P(AB) = P(B) - P(AB)$, 即 $P(\overline{AB}) = P(\overline{BA})$, 应选(C).

(8) 【答案】 (A).

【解】 因为 $X \sim N(\mu, \sigma^2)$, $Y \sim N(\mu, \sigma^2)$, 且 X, Y 相互独立,

所以 $X - Y \sim N(0, 2\sigma^2)$, 或 $\frac{X - Y}{\sqrt{2}\sigma} \sim N(0, 1)$,

故 $P\{|X - Y| < 1\} = P\left\{-\frac{1}{\sqrt{2}\sigma} < \frac{X - Y}{\sqrt{2}\sigma} < \frac{1}{\sqrt{2}\sigma}\right\} = 2\Phi\left(\frac{1}{\sqrt{2}\sigma}\right) - 1$,

即 $P\{|X - Y| < 1\}$ 与 μ 无关, 与 σ^2 有关, 应选(A).

二、填空题

(9) 【答案】 e^{-1} .

【解】
$$\lim_{n \rightarrow \infty} \left[\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \cdots + \frac{1}{n(n+1)} \right]^n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n+1} \right)^n$$
$$= \lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{n+1} \right)^{-(n+1)} \right]^{-\frac{n}{n+1}} = e^{-1}.$$

(10) 【答案】 $(\pi, -2)$.

【解】 $y' = x \cos x - \sin x$, $y'' = -x \sin x$,

令 $y'' = -x \sin x = 0$ 得 $x = 0, x = \pi$.

当 $x \in \left(-\frac{\pi}{2}, 0\right)$ 时, $y'' < 0$, 当 $x \in (0, \pi)$ 时, $y'' < 0$, 则点 $(0, 2)$ 不是拐点;

当 $x \in \left(\pi, \frac{3\pi}{2}\right)$ 时, $y'' > 0$, 故点 $(\pi, -2)$ 为拐点.

(11) 【答案】 $\frac{1 - 2\sqrt{2}}{18}$.

【解】

$$\begin{aligned} \int_0^1 x^2 f(x) dx &= \int_0^1 f(x) d\left(\frac{1}{3}x^3\right) = \frac{1}{3}x^3 f(x) \Big|_0^1 - \int_0^1 \frac{x^3}{3} \sqrt{1+x^4} dx \\ &= -\frac{1}{12} \int_0^1 \sqrt{1+x^4} d(1+x^4) = -\frac{1}{12} \times \frac{2}{3} (1+x^4)^{\frac{3}{2}} \Big|_0^1 = \frac{1 - 2\sqrt{2}}{18}. \end{aligned}$$

(12) 【答案】 0.4.

【解】 $\eta_{AA} = -\frac{dQ_A/dp_A}{Q_A/p_A} = (2p_A + p_B) \cdot \frac{p_A}{500 - p_A^2 - p_A p_B + 2p_B^2},$

当 $p_A = 10, p_B = 20$ 时, $\eta_{AA} = \frac{400}{500 - 100 - 200 + 800} = 0.4.$

(13) 【答案】 1.

【解】 因为 $\mathbf{A}\mathbf{X} = \mathbf{b}$ 有无穷多个解, 所以 $|\mathbf{A}| = 0$,

$$\text{而 } |\mathbf{A}| = \begin{vmatrix} 1 & 0 & -1 \\ 1 & 1 & -1 \\ 0 & 1 & a^2 - 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 1 & a^2 - 1 \end{vmatrix} = a^2 - 1, \text{ 则 } a = -1 \text{ 或 } a = 1.$$

当 $a = -1$ 时,

$$\bar{\mathbf{A}} = \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 1 & 1 & -1 & 1 \\ 0 & 1 & 0 & -1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & -2 \end{array} \right),$$

因为 $r(\mathbf{A}) \neq r(\bar{\mathbf{A}})$, 所以当 $a = -1$ 时, 方程组 $\mathbf{AX} = \mathbf{b}$ 无解;

当 $a = 1$ 时,

$$\bar{\mathbf{A}} = \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 1 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right),$$

因为 $r(\mathbf{A}) = r(\bar{\mathbf{A}}) = 2 < 3$, 所以当 $a = 1$ 时, 方程组 $\mathbf{AX} = \mathbf{b}$ 有无穷多个解, 故 $a = 1$.

(14) 【答案】 $\frac{2}{3}$.

$$\text{【解】 } E(X) = \int_0^2 x \cdot \frac{x}{2} dx = \frac{4}{3},$$

$$\text{分布函数 } F(x) = \int_{-\infty}^x f(x) dx,$$

当 $x < 0$ 时, $F(x) = 0$;

$$\text{当 } 0 \leq x < 2 \text{ 时, } F(x) = \int_0^x \frac{x}{2} dx = \frac{x^2}{4};$$

当 $x \geq 2$ 时, $F(x) = 1$, 即

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{x^2}{4}, & 0 \leq x < 2, \\ 1, & x \geq 2, \end{cases}$$

$$\text{故 } P\{F(X) > E(X) - 1\} = P\left\{F(X) > \frac{1}{3}\right\} = 1 - P\left\{F(X) \leq \frac{1}{3}\right\}$$

$$= 1 - P\left\{\frac{X^2}{4} \leq \frac{1}{3}\right\} = 1 - P\left\{X \leq \frac{2}{\sqrt{3}}\right\}$$

$$= 1 - \int_0^{\frac{2}{\sqrt{3}}} \frac{x}{2} dx = 1 - \frac{x^2}{4} \Big|_0^{\frac{2}{\sqrt{3}}} = \frac{2}{3}.$$

三、解答题

(15) 【解】 当 $x > 0$ 时, $f'(x) = (x^{2x})' = (e^{2x \ln x})' = x^{2x} \cdot (2 \ln x + 2)$;

当 $x < 0$ 时, $f'(x) = (x + 1)e^x$,

$$\text{由 } \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{e^{2x \ln x} - 1}{x} = \lim_{x \rightarrow 0^+} 2 \ln x = -\infty \text{ 得}$$

$f(x)$ 在 $x = 0$ 处不可导, 于是有

$$f'(x) = \begin{cases} x^{2x} (2 \ln x + 2), & x > 0, \\ (x + 1)e^x, & x < 0, \end{cases}$$

$f'(x)=0$ 或 $f(x)$ 的不可导点为 $x=-1, x=0, x=\frac{1}{e}$,

当 $x < -1$ 时, $f'(x) < 0$; 当 $-1 < x < 0$ 时, $f'(x) > 0$; 当 $0 < x < \frac{1}{e}$ 时, $f'(x) < 0$;

当 $x > \frac{1}{e}$ 时, $f'(x) > 0$,

故 $x=-1$ 为极小值点, 极小值为 $f(-1)=1-\frac{1}{e}$;

$x=0$ 为极大值点, 极大值为 $f(0)=1$;

$x=\frac{1}{e}$ 为极小值点, 极小值为 $f\left(\frac{1}{e}\right)=\left(\frac{1}{e}\right)^{\frac{2}{e}}$.

(16) 【解】 $\frac{\partial g}{\partial x}=y-f'_1-f'_2, \quad \frac{\partial g}{\partial y}=x-f'_1+f'_2,$

$$\frac{\partial^2 g}{\partial x^2} = -f''_{11} - 2f''_{12} - f''_{22},$$

$$\frac{\partial^2 g}{\partial x \partial y} = 1 - f''_{11} + f''_{12} - f''_{21} + f''_{22} = 1 - f''_{11} + f''_{22},$$

$$\frac{\partial^2 g}{\partial y^2} = -f''_{11} + f''_{12} + f''_{21} - f''_{22} = -f''_{11} + 2f''_{12} - f''_{22},$$

$$\begin{aligned} \text{故 } \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial x \partial y} + \frac{\partial^2 g}{\partial y^2} &= -f''_{11} - 2f''_{12} - f''_{22} + 1 - f''_{11} + f''_{22} - f''_{11} + 2f''_{12} - f''_{22} \\ &= 1 - 3f''_{11} - f''_{22}. \end{aligned}$$

(17) 【解】 (I) $y = \left(\int \frac{1}{2\sqrt{x}} e^{\frac{x^2}{2}} \cdot e^{\int -x dx} dx + C \right) e^{\int -x dx} = (\sqrt{x} + C) e^{\frac{x^2}{2}},$

由 $y(1)=\sqrt{e}$ 得 $C=0$, 即 $y(x)=\sqrt{x} e^{\frac{x^2}{2}}$.

$$(II) V = \int_1^2 \pi y^2 dx = \pi \int_1^2 x e^{x^2} dx = \frac{\pi}{2} e^{x^2} \Big|_1^2 = \frac{\pi}{2} (e^4 - e).$$

(18) 【解】 所求面积为

$$\begin{aligned} A &= \int_0^{+\infty} e^{-x} |\sin x| dx = \lim_{n \rightarrow \infty} \sum_{k=0}^n (-1)^k \int_{k\pi}^{(k+1)\pi} e^{-x} \sin x dx \\ &= \lim_{n \rightarrow \infty} \sum_{k=0}^n (-1)^k \left[-\frac{1}{2} e^{-x} (\sin x + \cos x) \right] \Big|_{k\pi}^{(k+1)\pi} \\ &= \frac{1}{2} \lim_{n \rightarrow \infty} \sum_{k=0}^n (-1)^{k+1} [e^{-(k+1)\pi} (-1)^{k+1} - e^{-k\pi} (-1)^k] \\ &= \frac{1}{2} \lim_{n \rightarrow \infty} \sum_{k=0}^n [e^{-(k+1)\pi} + e^{-k\pi}] = \frac{1}{2} \lim_{n \rightarrow \infty} \left[1 + 2 \sum_{k=1}^n e^{-k\pi} + e^{-(n+1)\pi} \right] \\ &= \frac{1}{2} \left(1 + 2 \sum_{k=1}^{\infty} e^{-k\pi} \right) = \frac{1}{2} \left(1 + \frac{2e^{-\pi}}{1 - e^{-\pi}} \right) = \frac{1}{2} \left(1 + \frac{2}{e^{\pi} - 1} \right) \\ &= \frac{1}{2} + \frac{1}{e^{\pi} - 1}. \end{aligned}$$

(19) (I)【证明】 因为当 $0 \leq x \leq 1$ 时, $x^{n+1} \sqrt{1-x^2} \leq x^n \sqrt{1-x^2}$,

所以 $\int_0^1 x^{n+1} \sqrt{1-x^2} dx < \int_0^1 x^n \sqrt{1-x^2} dx$, 即 $a_{n+1} < a_n$, 故 $\{a_n\}$ 单调递减.

$$a_n \stackrel{x=\sin t}{=} \int_0^{\frac{\pi}{2}} \sin^n t \cdot \cos^2 t dt = \int_0^{\frac{\pi}{2}} (\sin^n t - \sin^{n+2} t) dt$$

$$= \int_0^{\frac{\pi}{2}} \sin^n t dt - \int_0^{\frac{\pi}{2}} \sin^{n+2} t dt = I_n - \frac{n+1}{n+2} I_n = \frac{1}{n+2} I_n,$$

$$a_{n-2} \stackrel{x=\sin t}{=} \int_0^{\frac{\pi}{2}} \sin^{n-2} t \cdot \cos^2 t dt = \int_0^{\frac{\pi}{2}} (\sin^{n-2} t - \sin^n t) dt$$

$$= \int_0^{\frac{\pi}{2}} \sin^{n-2} t dt - \int_0^{\frac{\pi}{2}} \sin^n t dt = I_{n-2} - I_n,$$

因为 $I_n = \frac{n-1}{n} I_{n-2}$, 所以 $I_{n-2} = \frac{n}{n-1} I_n$,

于是 $a_{n-2} = \frac{n}{n-1} I_n - I_n = \frac{1}{n-1} I_n$,

故 $a_n = \frac{n-1}{n+2} a_{n-2} (n=2, 3, \dots)$.

(II)【解】 因为 $\{a_n\}$ 单调递减, 所以 $a_n = \frac{n-1}{n+2} a_{n-2} > \frac{n-1}{n+2} a_{n-1}$,

从而有 $\frac{n-1}{n+2} < \frac{a_n}{a_{n-1}} < 1$,

由夹逼定理得 $\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}} = 1$.

$$(20) \text{【解】 } (\alpha_1, \alpha_2, \alpha_3) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ 4 & 4 & a^2+3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & a^2-1 \end{pmatrix},$$

当 $a = -1$ 时, 向量组 $\alpha_1, \alpha_2, \alpha_3$ 的秩为 2,

$$\text{由 } (\beta_1, \beta_2, \beta_3) = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 3 \\ 2 & 2 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \text{ 得 } \beta_1, \beta_2, \beta_3 \text{ 的秩为 2,}$$

$$(\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3) = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 2 & 1 & 2 & 3 \\ 4 & 4 & 4 & 2 & 2 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & -2 & 2 & 0 \end{pmatrix},$$

因为 $r(\alpha_1, \alpha_2, \alpha_3) \neq r(\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3)$, 所以两个向量组不等价;

$$\text{当 } a = 1 \text{ 时, } (\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3) = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 2 & 1 & 2 & 3 \\ 4 & 4 & 4 & 4 & 0 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

因为 $r(\alpha_1, \alpha_2, \alpha_3) = r(\beta_1, \beta_2, \beta_3) = r(\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3) = 2$, 所以两个向量组等价,

令 $x_1 \alpha_1 + x_2 \alpha_2 + x_3 \alpha_3 = \beta_3$,

$$\text{再由 } (\alpha_1, \alpha_2, \alpha_3, \beta_3) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 2 & 3 \\ 4 & 4 & 4 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \text{ 得}$$

方程组 $x_1\alpha_1 + x_2\alpha_2 + x_3\alpha_3 = \beta_3$ 的通解为

$$\mathbf{X} = k \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -2 \\ 0 \end{pmatrix} = \begin{pmatrix} -2k+3 \\ k-2 \\ k \end{pmatrix} \quad (k \text{ 为任意常数}),$$

故 $\beta_3 = (-2k+3)\alpha_1 + (k-2)\alpha_2 + k\alpha_3$ (k 为任意常数).

当 $a \neq \pm 1$ 时, 向量组 $\alpha_1, \alpha_2, \alpha_3$ 的秩为 3,

$$\text{由 } (\beta_1, \beta_2, \beta_3) = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 3 \\ a+3 & 1-a & a^2+3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1-a & a^2-a \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & a^2-1 \end{pmatrix} \text{ 得}$$

向量组 $\beta_1, \beta_2, \beta_3$ 的秩为 3, 故两个向量组等价,

令 $x_1\alpha_1 + x_2\alpha_2 + x_3\alpha_3 = \beta_3$,

$$\begin{aligned} \text{由 } (\alpha_1, \alpha_2, \alpha_3, \beta_3) &= \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 0 & 2 & 3 \\ 4 & 4 & a^2+3 & a^2+3 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & a^2-1 & a^2-1 \end{array} \right) \\ &\rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array} \right) \text{ 得} \end{aligned}$$

$$\beta_3 = \alpha_1 - \alpha_2 + \alpha_3.$$

(21) 【解】 (I) 因为 $\mathbf{A} \sim \mathbf{B}$, 所以 $\text{tr } \mathbf{A} = \text{tr } \mathbf{B}$, 即 $x-4=y+1$, 或 $y=x-5$,

再由 $|\mathbf{A}| = |\mathbf{B}|$ 得 $-2(-2x+4) = -2y$, 即 $y = -2x+4$,

解得 $x=3, y=-2$.

$$\text{(II)} \mathbf{A} = \begin{pmatrix} -2 & -2 & 1 \\ 2 & 3 & -2 \\ 0 & 0 & -2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix},$$

显然矩阵 $\mathbf{A}, \mathbf{B}$ 的特征值都为 $\lambda_1 = -2, \lambda_2 = -1, \lambda_3 = 2$,

$$\text{由 } 2\mathbf{E} + \mathbf{A} \rightarrow \begin{pmatrix} 0 & -2 & 1 \\ 2 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & \frac{1}{4} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix} \text{ 得 } \mathbf{A} \text{ 的属于特征值 } \lambda_1 = -2 \text{ 的特征向量为}$$

$$\alpha_1 = \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix};$$

$$\text{由 } \mathbf{E} + \mathbf{A} \rightarrow \begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \text{ 得 } \mathbf{A} \text{ 的属于特征值 } \lambda_2 = -1 \text{ 的特征向量为}$$

$$\alpha_2 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix};$$

由 $2E - A \rightarrow \begin{pmatrix} 2 & 1 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ 得 A 的属于特征值 $\lambda_3 = 2$ 的特征向量为

$$\alpha_3 = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix},$$

令 $P_1 = \begin{pmatrix} -1 & -2 & -1 \\ 2 & 1 & 2 \\ 4 & 0 & 0 \end{pmatrix}$, 则 $P_1^{-1}AP_1 = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$;

由 $2E + B = \begin{pmatrix} 4 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ 得 B 的属于特征值 $\lambda_1 = -2$ 的特征向量为 $\beta_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$;

由 $E + B = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & \frac{1}{3} & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ 得 B 的属于特征值 $\lambda_2 = -1$ 的特征向量为

$$\beta_2 = \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix};$$

由 $2E - B = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ 得 B 的属于特征值 $\lambda_2 = 2$ 的特征向量为 $\beta_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$,

令 $P_2 = \begin{pmatrix} 0 & -1 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 0 \end{pmatrix}$, 则 $P_2^{-1}BP_2 = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

由 $P_1^{-1}AP_1 = P_2^{-1}BP_2$ 得 $(P_1P_2^{-1})^{-1}A(P_1P_2^{-1}) = B$,

故 $P = P_1P_2^{-1} = \begin{pmatrix} -1 & -1 & -1 \\ 2 & 1 & 2 \\ 0 & 0 & 4 \end{pmatrix}$.

(22) 【解】 (I) 因为 $X \sim E(1)$, 所以 X 的分布函数为

$$F(x) = \begin{cases} 1 - e^{-x}, & x \geq 0, \\ 0, & x < 0, \end{cases}$$

$$\begin{aligned} F_Z(z) &= P\{XY \leq z\} = P\{Y = -1\}P\{XY \leq z \mid Y = -1\} + P\{Y = 1\}P\{XY \leq z \mid Y = 1\} \\ &= pP\{-X \leq z\} + (1-p)P\{X \leq z\} = pP\{X \geq -z\} + (1-p)P\{X \leq z\} \\ &= p[1 - P\{X \leq -z\}] + (1-p)P\{X \leq z\} = p[1 - F(-z)] + (1-p)F(z), \end{aligned}$$

当 $z < 0$ 时, $F_Z(z) = pe^z$;

当 $z \geq 0$ 时, $F_Z(z) = p + (1-p)(1 - e^{-z})$,

故 $f_Z(z) = \begin{cases} pe^z, & z < 0, \\ (1-p)e^{-z}, & z \geq 0. \end{cases}$

$$\begin{aligned}(\text{II}) \operatorname{Cov}(X, Z) &= \operatorname{Cov}(X, XY) = E(X^2Y) - E(X) \cdot E(XY) \\ &= E(X^2)E(Y) - [E(X)]^2E(Y) = D(X) \cdot E(Y),\end{aligned}$$

因为 $X \sim E(1)$, 所以 $E(X) = 1, D(X) = 1$,

又因为 $Y \sim \begin{pmatrix} -1 & 1 \\ p & 1-p \end{pmatrix}$, 所以 $E(Y) = (-1)p + (1-p) = 1-2p$,

X 与 Z 不相关的充分必要条件是 $\operatorname{Cov}(X, Z) = 0$,

故当 $p = \frac{1}{2}$ 时, X 与 Z 不相关.

(III) 设 $F(x, z)$ 为 (X, Z) 的联合分布函数,

$$\begin{aligned}F(1, 1) &= P\{X \leq 1, Z \leq 1\} = P\{X \leq 1, XY \leq 1\} \\ &= P\{Y = -1\}P\{X \leq 1, XY \leq 1 \mid Y = -1\} + \\ &\quad P\{Y = 1\}P\{X \leq 1, XY \leq 1 \mid Y = 1\} \\ &= pP\{X \leq 1, -X \leq 1\} + (1-p)P\{X \leq 1\} \\ &= pP\{-1 \leq X \leq 1\} + (1-p)P\{X \leq 1\} \\ &= P\{X \leq 1\} = F(1) = 1 - \frac{1}{e},\end{aligned}$$

$$F_X(1) = P\{X \leq 1\} = 1 - \frac{1}{e},$$

$$F_Z(1) = p + (1-p)\left(1 - \frac{1}{e}\right),$$

因为 $F(1, 1) \neq F_X(1) \cdot F_Z(1)$, 所以 X 与 Z 不相互独立.

(23) 【解】 (I) 由归一性得

$$\begin{aligned}1 &= \int_{\mu}^{+\infty} \frac{A}{\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = A \int_{\mu}^{+\infty} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} d\left(\frac{x-\mu}{\sigma}\right) = A \int_0^{+\infty} e^{-\frac{t^2}{2}} dt \\ &= \sqrt{2\pi} A \int_0^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = \frac{\sqrt{2\pi}}{2} A \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = \frac{\sqrt{2\pi}}{2} A,\end{aligned}$$

解得 $A = \sqrt{\frac{2}{\pi}}$.

$$(\text{II}) \text{ 似然函数为 } L(\sigma^2) = \frac{A^n}{(\sigma^2)^{\frac{n}{2}}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2},$$

$$\text{取对数 } \ln L(\sigma^2) = n \ln A - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2,$$

$$\text{由 } \frac{d}{d\sigma^2} \ln L(\sigma^2) = -\frac{n}{2} \cdot \frac{1}{\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 = 0 \text{ 得}$$

$$\sigma^2 \text{ 的最大似然估计值为 } \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2,$$

$$\text{故 } \sigma^2 \text{ 的最大似然估计量为 } \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2.$$